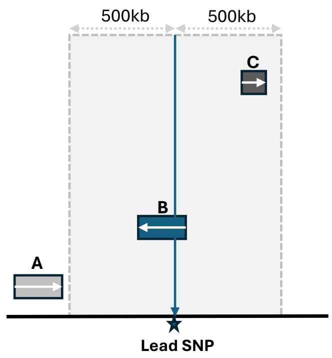
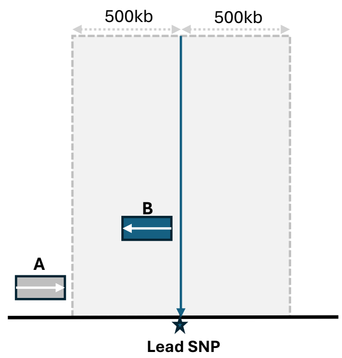
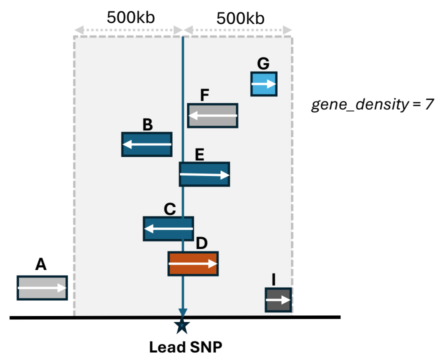
i) Single genic: ***nearest_gene* = *B***



ii) Non-genic single flanking: ***nearest_gene* = *B***



iii) Multiple genic: ***nearest_gene* = *prioritized***

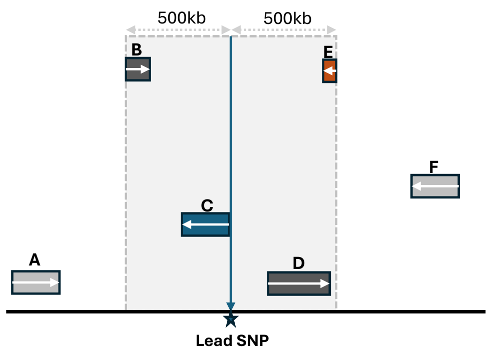


Multiple genic nearest gene prioritization

- ***genic***: True (1), False (0)
- ***promoter*** (is SNP within 2 kb upstream of gene TSS?): True (1), False (0)
- ***biotype_weight***: See BIOTYPE_WEIGHTS
- ***distance_score*** (weighted - smaller distance higher weight): $\frac{1}{log10\ (distance+10)}$

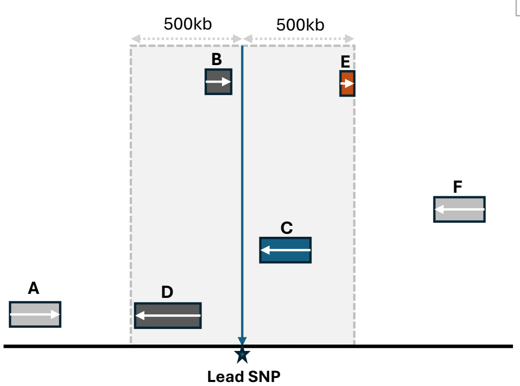
$$\boldsymbol{priority\_score} = \underbrace{2(\boldsymbol{genic})}_{\text{Ensures genic candidates are prioritized whether protein-coding or not.}} + 1(\boldsymbol{promoter}) + \underbrace{2(\boldsymbol{biotype\_weight})}_{\text{When all candidates are either only genic or non-genic, ensures protein-coding/gene are prioritized.}}\ \boldsymbol{x\ distance\_score}$$

iv) Intergenic: ***nearest_downstream_gene* = *C***

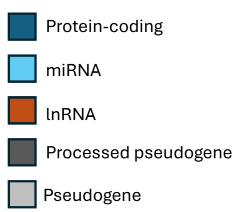


When intergenic, pycmplot ensures that one gene is always selected from either side of the SNP based on distance to create a top_gene field LEFT_GENE—RIGHT_GENE join (e.g., HBS1L-MYB)

v) Intergenic: ***nearest_upstream_gene* = *B***



top_gene* = *LEFT_GENE—RIGHT_GENE



BIOTYPE_WEIGHTS
(pycmplot/constants.py)

gene / protein_coding	1
IG / TR gene	1
protein_coding_LoF	0.90
protein_coding_CDS_not_defined	0.85
miRNA	0.75
piRNA / siRNA / ribozyme	0.70
lncRNA / lincRNA / ncRNA	0.70
snRNA / snoRNA / scaRNA	0.65
antisense / macro_lncRNA	0.65
vault_RNA / vaultRNA / tRNA	0.60
sense_intronic / sense_overlapping	0.60
nonsense_mediated_decay	0.55
rRNA / miscRNA / retained_intron	0.55
processed_transcript	0.50
transcribed_processed_pseudogene	0.45
transcribed_unitary_pseudogene	0.40
transcribed_unprocessed_pseudogene	0.35
readthrough	0.35
processed / polymorphic pseudogene	0.30
unitary_pseudogene	0.25
pseudogene / unprocessed_pseudogene / IG / TR	0.20